

# P09 Hq Solution

## Origin of Matter-Antimatter Asymmetry (HQ v54)

**Domain:** Particle Physics/Cosmology

**Model:** PAX L5 Narrow L2 General 27B

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**Run:** PAX\_27B HQ v54 Tokens: 7399 Time: 1292.1s Chain:

7c2a10f275308257b95335f6839c1565159650e399175675822ed873083c14f7 Genesis:

8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560

**Subject:** Proposed Resolution to the Baryon Asymmetry Problem via Resonant Leptogenesis with CP-Violating Neutrino Oscillations in the Early Universe

### ### 1. Core Hypothesis

The observed baryon asymmetry of the universe (BAU), quantified by  $\eta_B \approx 6 \times 10^{-10}$ , originates from **Resonant Leptogenesis** driven by the decay of heavy right-handed Majorana neutrinos ( $N_i$ ) in the early universe. Unlike standard thermal leptogenesis which requires extremely high mass scales ( $M_N > 10^9$  GeV) and often conflicts with low-energy neutrino data, this hypothesis posits that the necessary CP violation arises from **quasi-degenerate masses** of the lightest two right-handed neutrinos ( $M_1 \approx M_2$ ). This degeneracy enhances the CP-asymmetry parameter  $\epsilon_{CP}$  through resonant self-energy corrections, allowing successful baryogenesis at accessible energy scales ( $M_N \sim 10$  TeV), potentially within reach of future colliders or precision intensity frontier experiments. The generated lepton asymmetry is subsequently converted into baryon asymmetry via non-perturbative electroweak sphaleron processes before they freeze out.

### ### 2. Mathematical Framework

The framework relies on the extension of the Standard Model (SM) by three right-handed singlet fermions  $N_R$ , leading to the Type-I Seesaw mechanism.

#### A. Lagrangian Extension:

$$\mathcal{L}_{\text{BSM}} = \mathcal{L}_{\text{SM}} - \frac{1}{2} \overline{N_R^c} M_R N_R - \left( \overline{L_L} Y_\nu \tilde{\Phi} N_R + \text{h.c.} \right)$$

Where:

\*  $L_L$ : SM lepton doublet.

\*  $\Phi$ : Higgs doublet,  $\tilde{\Phi} = i\sigma_2 \Phi^*$ .

\*  $Y_\nu$ : Yukawa coupling matrix.

\*  $M_R$ : Majorana mass matrix for  $N_R$ .

After Electroweak Symmetry Breaking (EWSB), the light neutrino mass matrix is given by the seesaw formula:

$$m_\nu \approx -v^2 Y_\nu M_R^{-1} Y_\nu^T$$

#### B. Resonant Enhancement of CP Asymmetry:

In the limit where  $M_1 - M_2 \lesssim \Gamma_{1,2}/2$  (where  $\Gamma$  is the decay width), the CP asymmetry parameter  $\epsilon_1$  for the decay  $N_1 \rightarrow \ell \phi$  is enhanced beyond the standard perturbative bound. The total asymmetry includes vertex and self-energy contributions:

$$\epsilon_1 = \frac{1}{8\pi} (Y_\nu^\dagger Y_\nu)_{11} \sum_{j \neq 1} \text{Im} \left[ (Y_\nu^\dagger Y_\nu)_{1j}^2 \right] f(M_j^2/M_1^2)$$

For quasi-degenerate states ( $M_1 \approx M_2$ ), the loop function  $f(x)$  becomes dominated by the Breit-Wigner resonance:

$$\epsilon_1 \approx \frac{1}{8\pi} \frac{(Y_\nu^\dagger Y_\nu)_{12} \text{Im}(Y_\nu^\dagger Y_\nu)_{22}}{(Y_\nu^\dagger Y_\nu)_{12}^2} \frac{1}{\Gamma_2} \frac{1}{(M_1 - M_2)^2 + (\Gamma_2/2)^2}$$

This allows  $\epsilon_1$  to approach  $\mathcal{O}(1)$  even if Yukawa couplings are small, satisfying the Sakharov conditions efficiently.

### C. Boltzmann Evolution:

The evolution of the number densities  $n_{\{N_i\}}$  and the lepton asymmetry  $n_{\{L-\alpha\}}$  is governed by coupled Boltzmann equations including washout effects:

$$\frac{dn_{\{N_i\}}}{dz} = -D_i(z) \left( \frac{n_{\{N_i\}}}{n_{\{N_i\}}^{eq}} - 1 \right)$$

$$\frac{dn_{\{L-\alpha\}}}{dz} = -\sum_i \epsilon_{\{i,\alpha\}} D_i(z) \left( \frac{n_{\{N_i\}}}{n_{\{N_i\}}^{eq}} - 1 \right) - W_{\alpha}(z) n_{\{L-\alpha\}}$$

Where  $z = T/M_1$ ,  $D_i$  represents the decay rate, and  $W_{\alpha}$  represents the inverse decay and scattering washout terms.

#### ### 3. Proof Sketch / Key Steps

##### 1. Sakharov Condition Verification:

\* **Baryon Number Violation**: Provided by non-perturbative sphalerons in the SM plasma above  $T_c \approx 130$  GeV.

\* **C and CP Violation**: Generated by complex phases in the Yukawa matrix  $Y_{\nu}$  during  $N_i$  decays, amplified by the resonant denominator  $(M_1 - M_2)^2 + \Gamma^2/4$ .

\* **Departure from Thermal Equilibrium**: Achieved when the decay rate  $\Gamma_{N_1} < H(T=M_1)$  (Hubble expansion rate), ensuring the process occurs out-of-equilibrium.

##### 2. Derivation of the Resonance Condition:

\* Demonstrate that for hierarchical masses,  $\epsilon_1 \propto M_1$ . To match  $\eta_B$ , one needs  $M_1 > 10^9$  GeV.

\* Show mathematically that introducing a splitting  $\Delta M = M_1 - M_2$  such that  $\Delta M \sim \Gamma$  creates a pole in the propagator correction, boosting  $\epsilon_1$  by a factor of  $M/\Gamma$ .

\* Calculate the required fine-tuning:  $\Delta M / M \sim 10^{-8}$  to  $10^{-12}$  depending on the specific Yukawa texture, arguing this is natural in models with approximate flavor symmetries (e.g.,  $SU(2)_R$ ).

##### 3. Flavor Effects Integration:

\* Incorporate "flavored" Boltzmann equations. At temperatures  $T < 10^9$  GeV, charged lepton Yukawa interactions maintain coherence between flavors.

\* Prove that flavor correlations can suppress washout rates for specific lepton flavors while preserving asymmetry in others, significantly relaxing the constraints on  $M_{N_i}$ .

##### 4. Conversion to Baryon Asymmetry:

\* Apply the sphaleron conversion factor:  $n_B = c_{\text{sph}} n_L$ . In the SM,  $c_{\text{sph}} \approx -28/79$ .

\* Integrate the final lepton asymmetry  $Y_L$  over the temperature history to yield the final baryon-to-photon ratio  $\eta_B = n_B/n_{\gamma}$ .

#### ### 4. Testable Predictions

If this hypothesis holds, it yields distinct experimental signatures distinguishable from high-scale leptogenesis:

##### 1. Low-Scale Heavy Neutral Leptons (HNLs):

\* Existence of sterile neutrinos with masses in the MeV–TeV range.

\* **Search Strategy**: Displaced vertices at LHCb, ATLAS, CMS, or dedicated experiments like SHiP, FASER, and DUNE near detectors.

##### 2. Lepton Flavor Universality Violation (LFUV):

\* Specific patterns in rare meson decays (e.g.,  $K \rightarrow \pi \nu \bar{\nu}$ ,  $B \rightarrow K \mu e$ ) induced by mixing between active and sterile neutrinos.

##### 3. Neutrinoless Double Beta Decay ( $0\nu\beta\beta$ ):

\* While standard seesaw predicts very long half-lives due to tiny  $m_{\nu}$ , the low-scale resonant scenario may allow observable rates if the effective Majorana mass  $m_{ee}$  receives contributions from the heavy sector interference or if the light neutrino spectrum is inverted/quasi-degenerate.

##### 4. Cosmic Microwave Background (CMB) Constraints:

\* Precise measurements of the effective number of relativistic species ( $N_{\text{eff}}$ ) and primordial helium abundance ( $Y_p$ ) could constrain the lifetime and decay modes of these HNLs in the early universe.

##### 5. Gravitational Waves:

\* If the phase transition associated with the generation of asymmetry involves a strong first-order phase transition (linked to the symmetry breaking scale), it would generate a stochastic gravitational wave background detectable by LISA or DECIGO.

### ### 5. Remaining Open Sub-Problems

Despite the theoretical elegance, several critical challenges remain:

#### 1. **Fine-Tuning vs. Naturalness:**

\* Achieving the precise mass degeneracy ( $\Delta M \sim \Gamma$ ) required for resonance enhancement without an underlying protective symmetry remains difficult. Is there a dynamical mechanism (e.g., radiative stability) that enforces this degeneracy?

#### 2. **Thermal Corrections and Plasma Effects:**

\* Accurate calculation of the thermal widths  $\Gamma(T)$  and thermal masses in the dense early universe plasma is computationally intensive. Current approximations may underestimate washout effects or shift the resonance condition.

#### 3. **Connection to Low-Energy Neutrino Data:**

\* Mapping the high-energy parameters ( $Y_\nu, M_R$ ) to the measured low-energy observables ( $m_\nu$ , PMNS mixing angles, Dirac CP phase  $\delta_{CP}$ ) is highly non-trivial. There is a vast parameter space; determining if the specific textures required for resonant leptogenesis are compatible with current oscillation data (especially the normal vs. inverted hierarchy preference) is ongoing.

#### 4. **Non-Perturbative Sphaleron Dynamics:**

\* The exact efficiency of the sphaleron conversion depends on the detailed dynamics of the electroweak phase transition. If the transition is a smooth crossover (as suggested by lattice QCD for the SM), the window for converting lepton asymmetry to baryon asymmetry narrows, requiring extensions to the scalar sector (e.g., Two-Higgs-Doublet Models) to ensure a strong first-order phase transition.

#### 5. **Initial Conditions:**

\* The solution assumes a thermal initial state. However, inflationary reheating scenarios might produce non-thermal distributions of  $N_i$ , altering the Boltzmann evolution entirely. A robust theory must be independent of specific reheating histories.